

7	Answer: A
	The work done by the net force provides the change in kinetic energy, which is equal to the kinetic energy if the initial kinetic energy is 0. KE at displacement s = work done by the net force = $\int F ds$ = area under the F-s graph Hence, gradient of KE-s graph = F. (It's easier to work with gradient.) From s_1 to s_2, F increases from 0, so the gradient of KE-s increases from 0. From s_2 to s_3, F decreases from a positive value to 0, so the gradient of KE-s decreases. Only A fits the above description. KE is always positive, so reject options C and D. Area under F-s graph = work done by F = Gain in KE. From s_1 to s_3, area always increasing, also F always positive, so KE always increasing, thus reject option B.

8	Answer: C
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